(Viewing data as-entered. Click here to return)

AK Steel 17-4 PH® Precipitation Hardening Stainless Steel, Condition H 900

Categories: Metal; Ferrous Metal; Martensitic; Stainless Steel; Precipitation Hardening Stainless

Material Notes: AK Steel 17-4 PH® provides an outstanding combination of high strength, good corrosion resistance, good mechanical properties at temperatures up to 600°F (316°C), and good toughness in both base metal and welds. Short-time, low-temperature heat treatments minimize distortion and scaling. This alloy is widely used in the aerospace, chemical, petrochemical, food processing, paper and general metalworking industries.

The material supplied from the mill is in Condition A. After fabrication eight standard heat treatments have been developed to provide a wide range of properties.

Information provided by AK Steel

Key Words: UNS S17400

Vendors: **Click here to view all available suppliers for this material.**

Please click here if you are a supplier and would like information on how to add your listing to this material.

Printer friendly version Download as PDF Download to Excel (requires Excel and Windows)
Export data to your CAD/FEA program

Add to Folder: My Folder 0/0

Physical Properties	Original Value	Comments
Density	7.80 g/cc	

Mechanical Properties	Original Value	Comments
Hardness, Rockwell C	45	
Tensile Strength, Ultimate	1448 MPa	
Tensile Strength, Yield	1379 MPa @Strain 0.2 %	
Elongation at Break	7.0 %	in 2 inches

Electrical Properties	Original Value	Comments
Electrical Resistivity	0.0000770 ohm-cm	

Thermal Properties	Original Value	Comments
CTE, linear	10.8 µm/m-°C @Temperature 21.0 - 93.0 °C	
	11.7 µm/m-°C @Temperature <=427 °C	
Specific Heat Capacity	0.460 J/g-°C	

Thermal Conductivity	17.9 W/m-K @Temperature 149 °C	
	22.6 W/m-K @Temperature 482 °C	

Component Elements Properties	Original Value	Comments
Carbon, C	<= 0.070 %	
Chromium, Cr	15 - 17.5 %	
Copper, Cu	3.0 - 5.0 %	
Iron, Fe	69.91 - 78.85 %	As Remainder
Manganese, Mn	<= 1.0 %	
Nb + Ta	0.15 - 0.45 %	
Nickel, Ni	3.0 - 5.0 %	
Phosphorus, P	<= 0.040 %	
Silicon, Si	<= 1.0 %	
Sulfur, S	<= 0.030 %	

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's terms of use regarding this information. Click here to go back to viewing the property data in MatWeb's normal format.

Users viewing this material also viewed the following:

AK Steel 17-4 PH® Precipitation Hardening Stainless Steel, Condition A
Kinetics MIM 17-4PH Stainless Steel (H900 Condition)
AK Steel 17-4 PH® Precipitation Hardening Stainless Steel, Condition H 1025
AK Steel 17-4 PH® Precipitation Hardening Stainless Steel, Condition H 925
Carpenter Custom 630 (17-Cr-4Ni) Precipitation Hardening Stainless Steel, Condition H 900 (Age-Hardened 482°C/1hr/AC)

NAKS129 / 14732